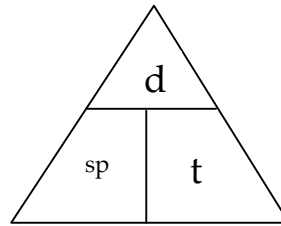


**YEAR 10 SCIENCE
PHYSICAL SCIENCES
PHYSICS DATA SHEET**

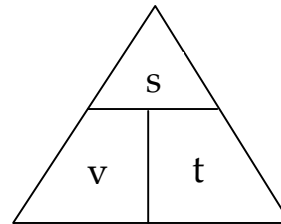
Speed

$$\text{speed} = \frac{d}{t}$$



Velocity

$$v = \frac{s}{t}$$



Acceleration

$$a = \frac{v - u}{t}$$

Equations of motion

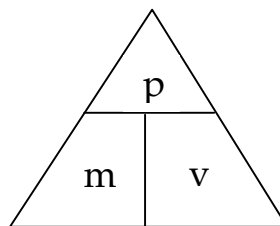
$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

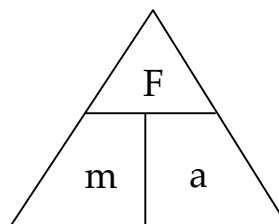
Momentum

$$p = mv$$



Force

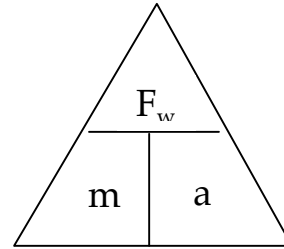
$$F = ma$$



$$F = \frac{mv - mu}{t}$$

Weight

$$F_w = mg$$



Acceleration due to gravity

$$g = 9.80 \text{ ms}^{-2}$$

Converting kmh^{-1} to ms^{-1}

Divide the speed in kmh^{-1} by 3.6 to get the speed in ms^{-1} .